



Основы глубокого обучения, 2-е издание

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Тип файла: pdf

Размер: 15.9 MB

Язык: Английский

Страниц: 390

Основы глубокого обучения 2-е издание: Полное руководство по нейронным сетям, искусственному интеллекту и практическим приложениям глубокого обучения 🤖📖

Введение 🚀

Искусственный интеллект изменил почти все инженерные дисциплины, от робототехники и здравоохранения до автономных транспортных средств и промышленной автоматизации. В центре этой революции находится **глубокое обучение** — раздел машинного обучения, который позволяет компьютерам выявлять сложные закономерности в огромных массивах данных.

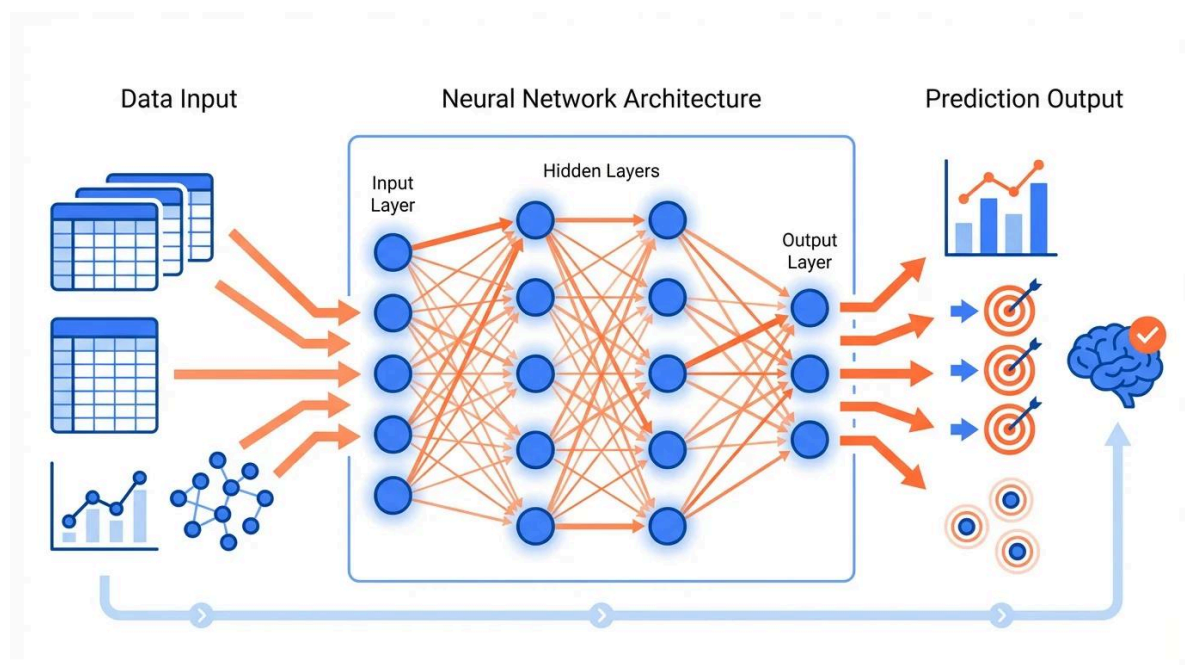
Одним из самых авторитетных образовательных ресурсов на сегодняшний день является книга «**Основы глубокого обучения**», **2-е издание**, написанная для того, чтобы помочь начинающим и опытным инженерам разобраться как в математических концепциях, так и в практической реализации систем глубокого обучения.

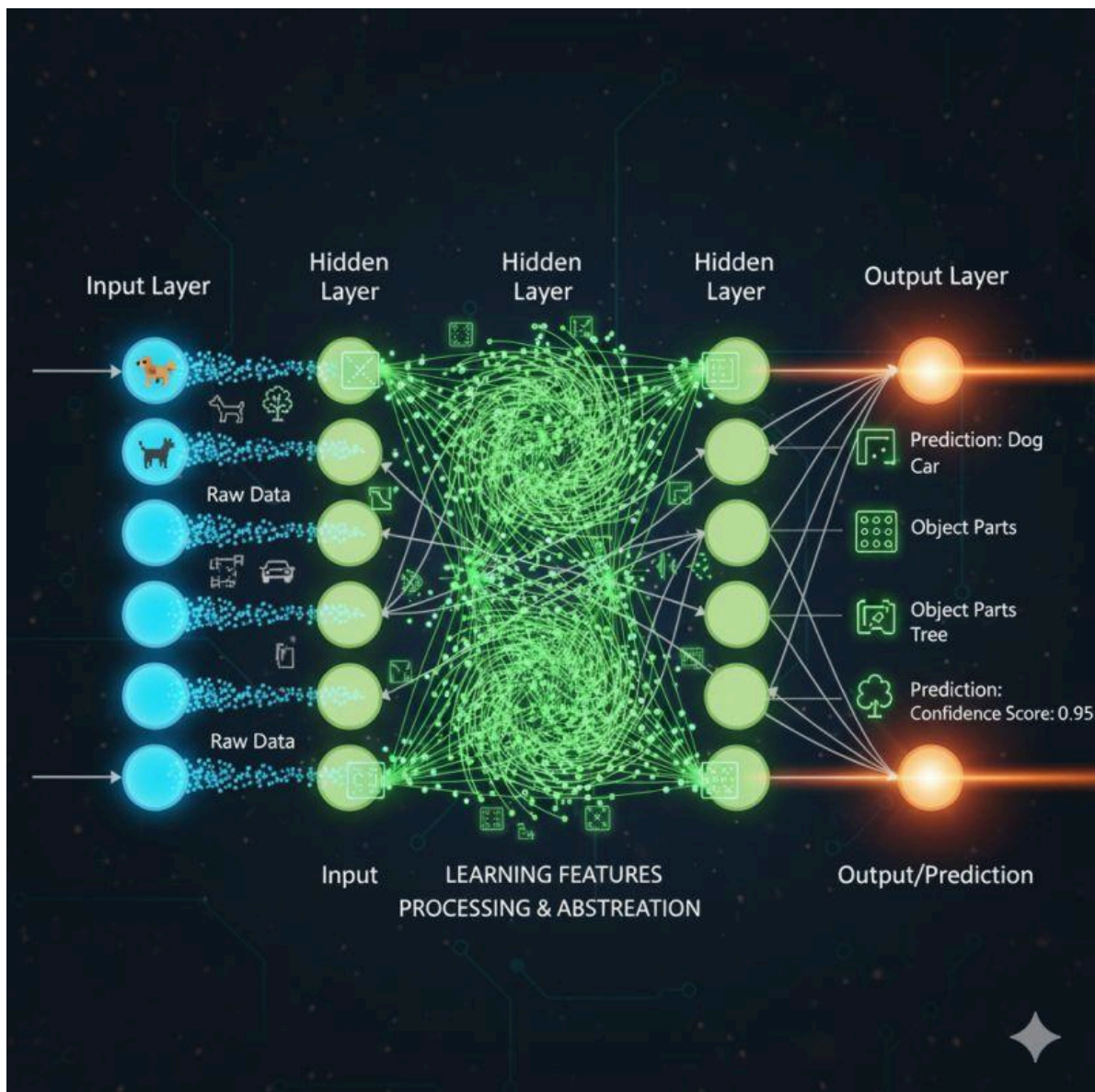
В отличие от традиционного программирования, где разработчики явно определяют каждое правило, глубокое обучение позволяет компьютерам автоматически выявлять взаимосвязи в процессе обучения. Эта возможность лежит в основе таких технологий, как:

- 🤖 Интеллектуальные роботы
- 🚗 Беспилотные автомобили
- 🏥 Медицинская диагностика
- 📷 Компьютерное зрение

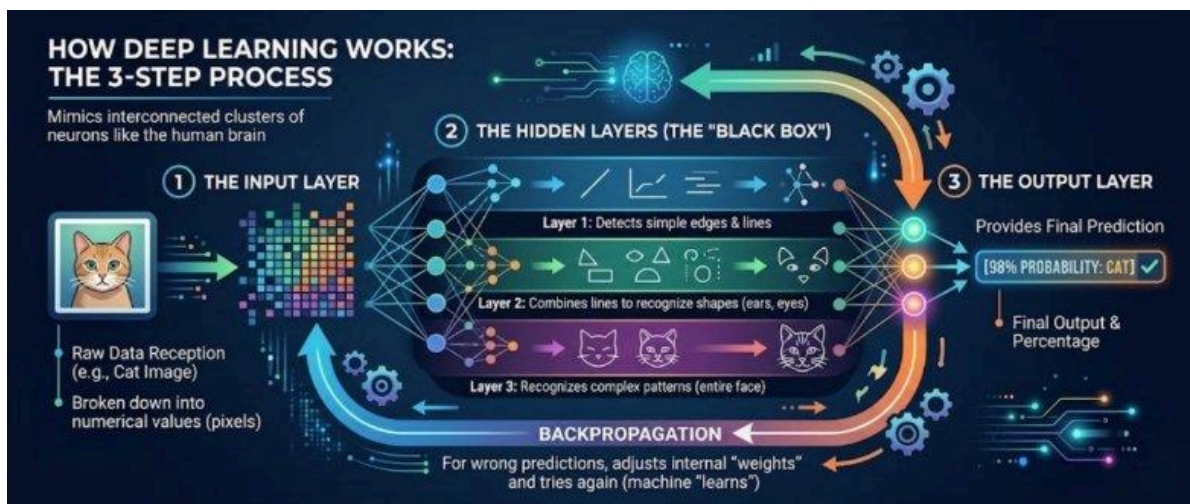
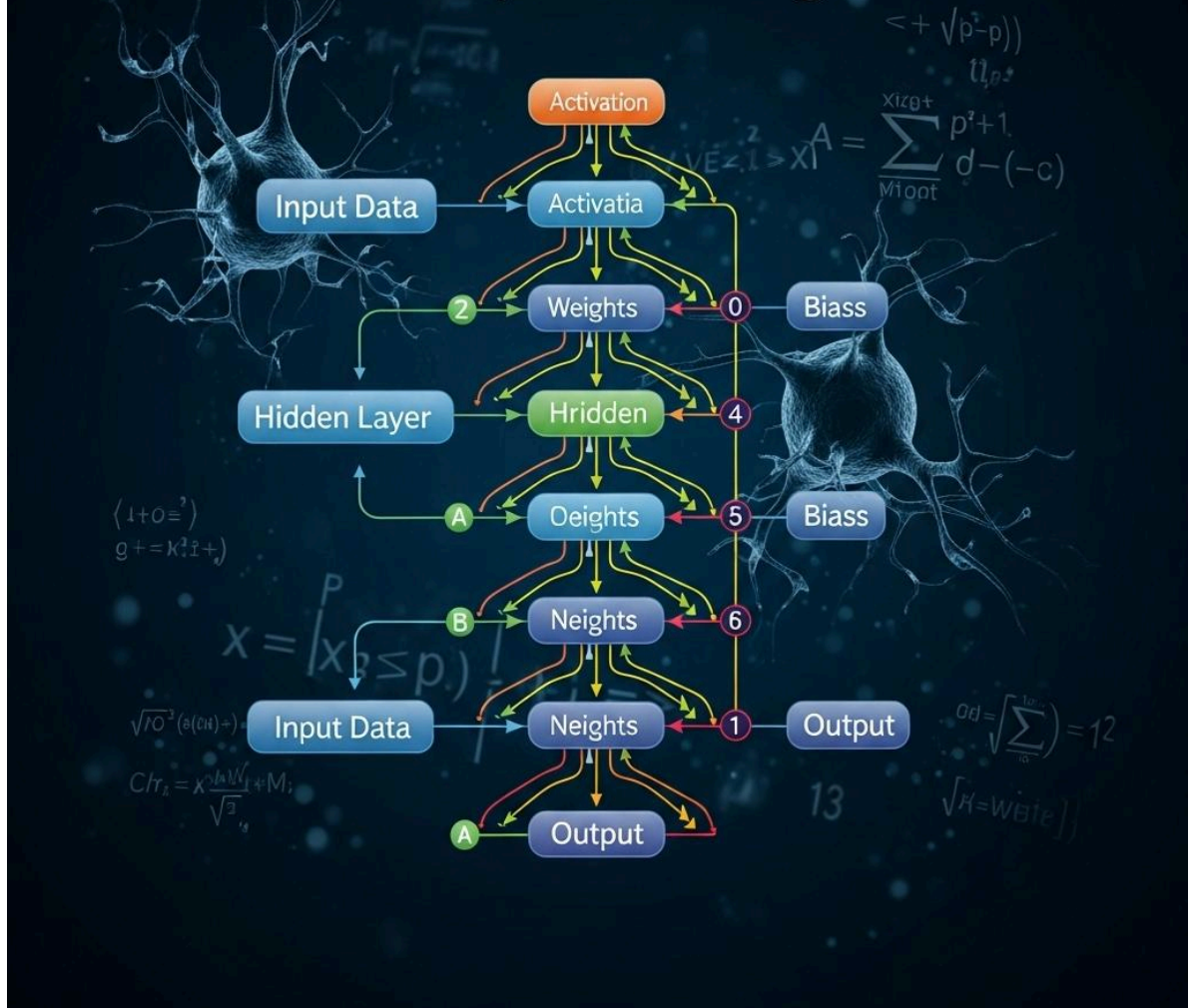
- 🗣️ Распознавание речи
- 🌐 Языковой перевод
- 💬 Чат-боты с искусственным интеллектом
- 📊 Предиктивная аналитика

В этом руководстве рассматриваются основные идеи, представленные в книге «Основы глубокого обучения», 2-е издание, чтобы сделать их доступными для студентов инженерных специальностей, исследователей, разработчиков программного обеспечения и специалистов в области искусственного интеллекта из США, Великобритании, Канады, Австралии и Европы.





Deep Learning



Теоретическая основа

Глубокое обучение основано на десятилетиях исследований в области:

Искусственного интеллекта

Искусственный интеллект направлен на создание машин, способных выполнять задачи, которые обычно требуют человеческого интеллекта.

Примеры:

- Принятие решений
- Распознавание образов
- Планирование
- Понимание языка

Машинное обучение

Машинное обучение — это раздел искусственного интеллекта, в рамках которого компьютеры обучаются на основе данных, а не с помощью явного программирования.

Вместо того чтобы писать:

```
1 | IF image contains whiskers  
2 | THEN cat
```

Модель машинного обучения обучается этому автоматически после анализа тысяч размеченных изображений.

Нейронные сети

Нейронные сети имитируют структуру биологических нейронов.

Сеть обычно состоит из:

- Входного слоя
- Скрытых слоев
- Выходного слоя

Каждый нейрон выполняет математические операции и передает информацию на следующий слой.

Чем глубже сеть, тем более сложные закономерности она может распознавать.

Почему глубокое обучение стало успешным

Этому способствовали несколько технологических достижений:

✓ Более быстрые графические процессоры

✓ Большие наборы данных

 Более совершенные алгоритмы

✓ Облачные вычисления

✓ Фреймворки с открытым исходным кодом

Примеры:

- TensorFlow
- PyTorch
- Keras
- JAX

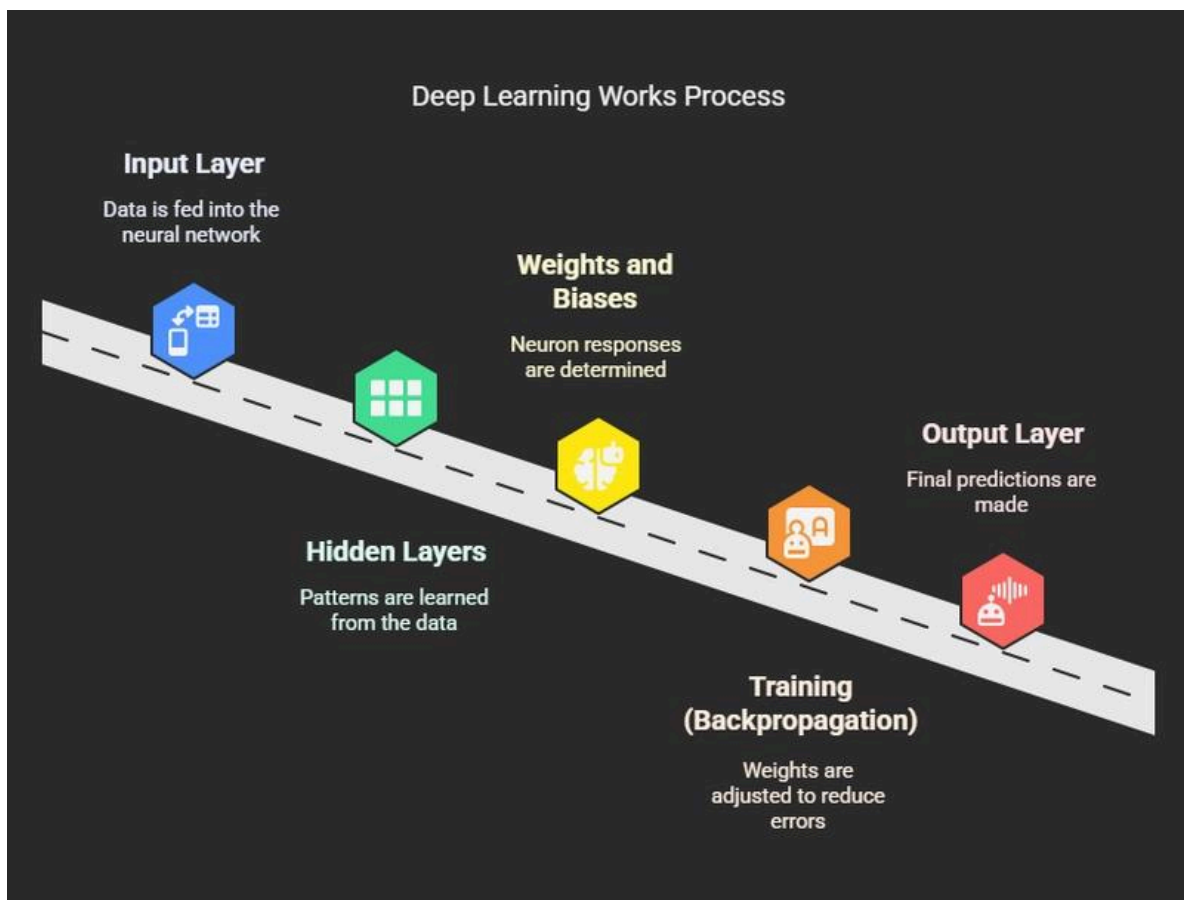
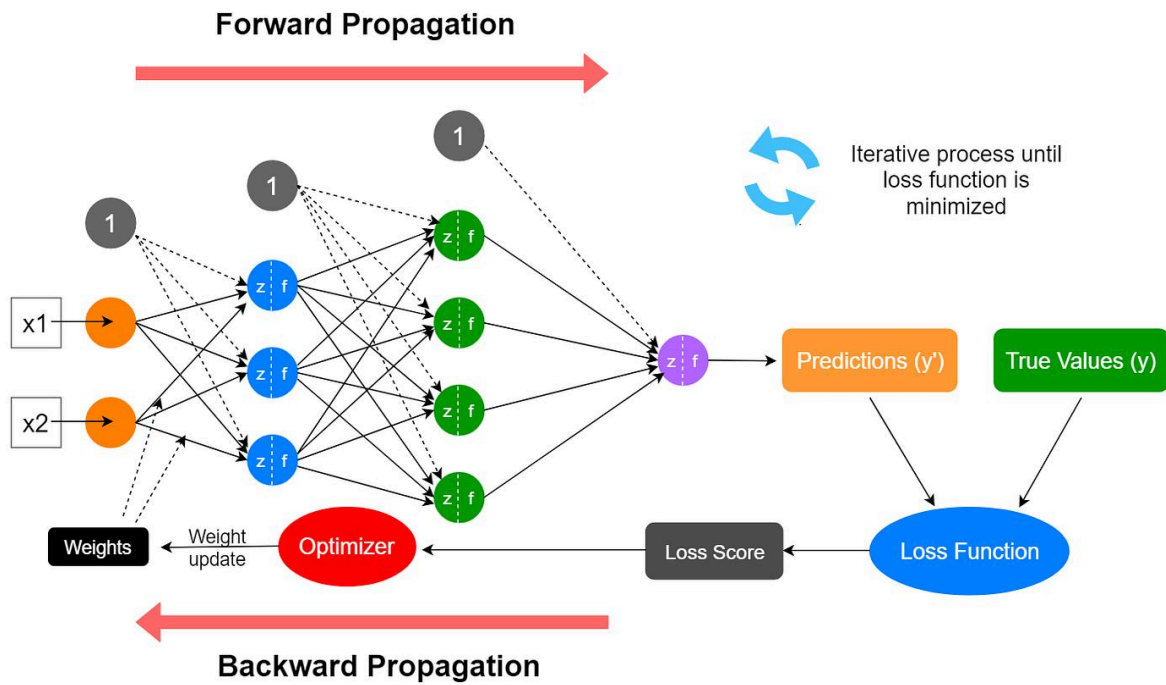
Definition

Fundamentals of Deep Learning, 2nd Edition is an educational resource that teaches engineers how modern neural networks function, how to train them efficiently, and how to apply deep learning techniques to real-world engineering problems.

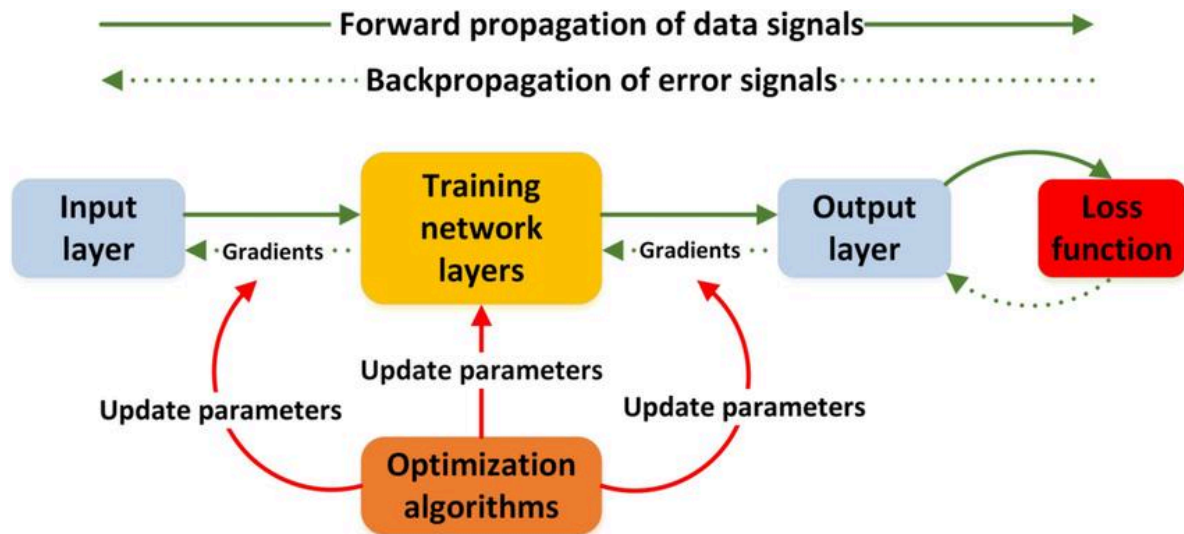
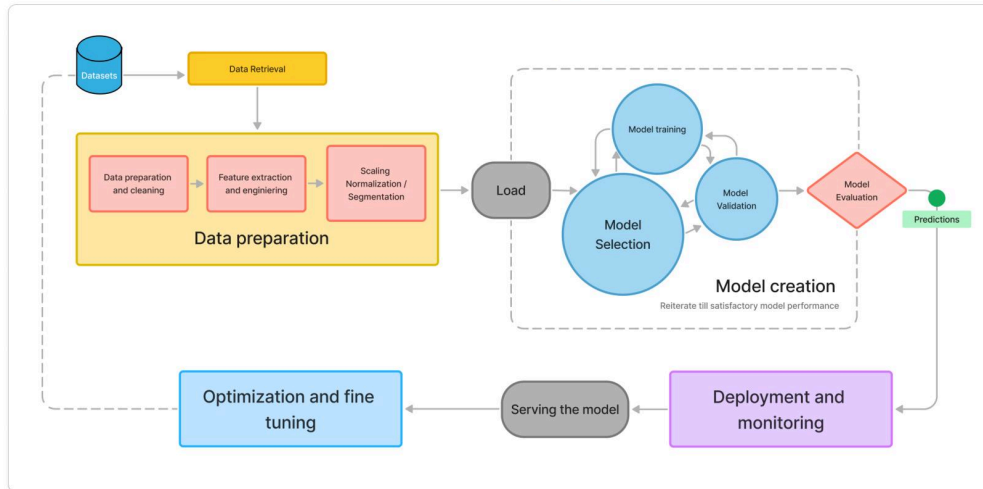
The book combines:

- Mathematical foundations
- Practical programming
- GPU computing
- Neural network architectures
- Computer vision
- Natural Language Processing
- Reinforcement Learning
- Production deployment

Step-by-Step Explanation



Machine learning pipeline



Step 1 — Collect Data 📥

Every deep learning project begins with data.

Examples:

- Images
- Videos
- Audio
- Sensor readings
- Medical records
- Financial transactions

More data generally leads to better performance.

Step 2 — Clean the Dataset 🖌️

Raw data often contains:

- Missing values
- Duplicate entries
- Incorrect labels
- Noise

Cleaning improves training quality.

Step 3 — Prepare Features

Engineers often normalize data:

- Scale numerical values
- Resize images
- Convert text into vectors

Proper preprocessing greatly improves learning.

Step 4 — Build the Neural Network

Design decisions include:

- Number of hidden layers
- Activation functions
- Optimizer
- Learning rate
- Batch size

Step 5 — Train the Model

Training adjusts millions of weights using:

- Forward propagation
- Loss calculation
- Backpropagation
- Gradient descent

The process repeats across many epochs.

Step 6 — Validate Performance

Metrics include:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC Curve

Step 7 — Deploy the Model

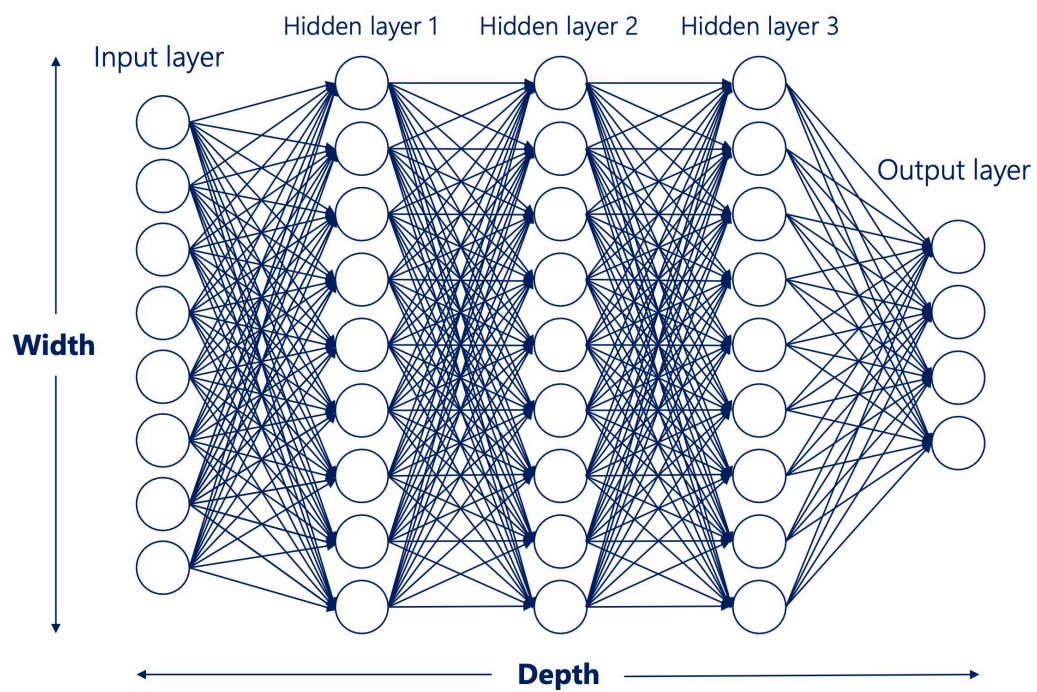
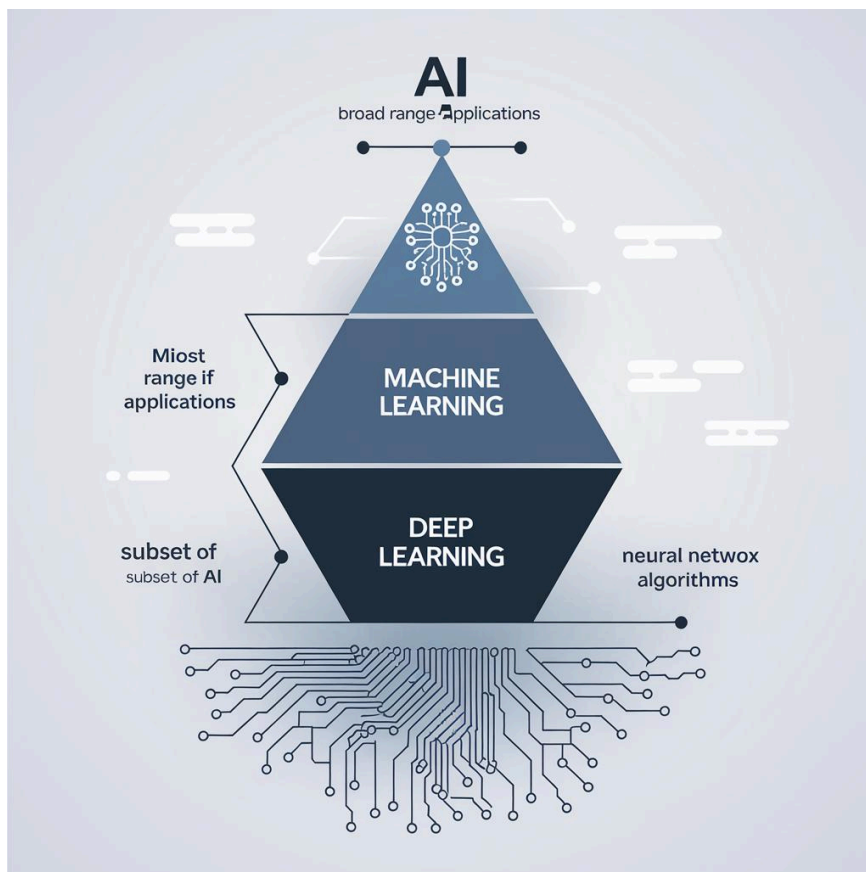
After successful testing, engineers deploy models into:

- Mobile apps
- Web applications
- Embedded systems
- Cloud services

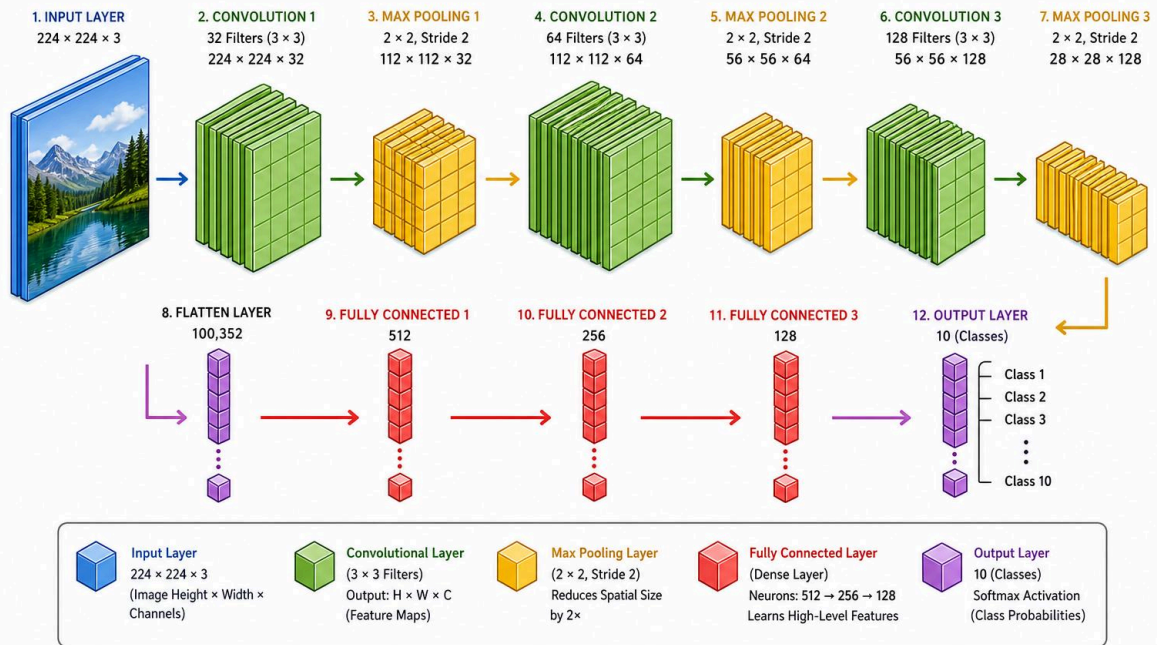
Comparison

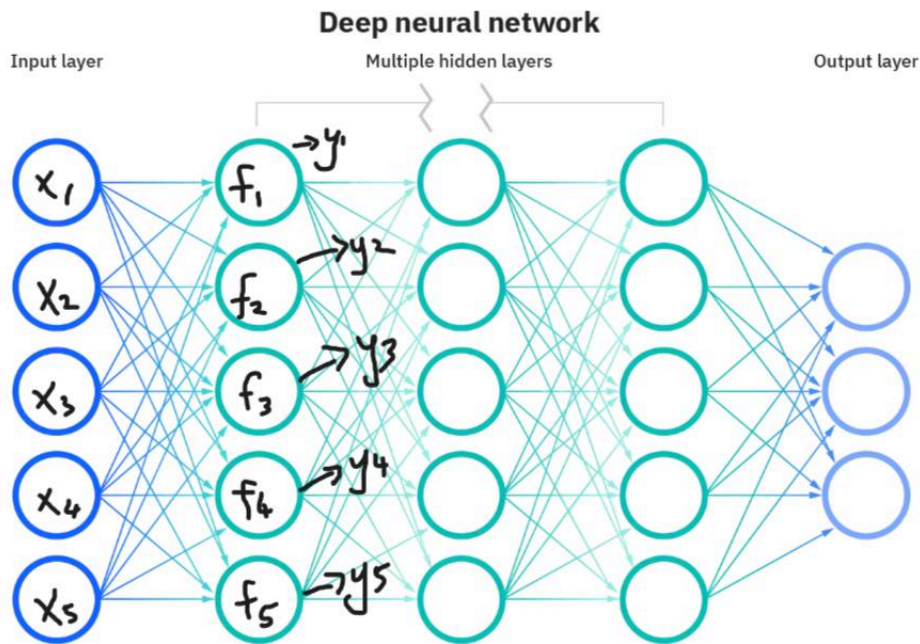
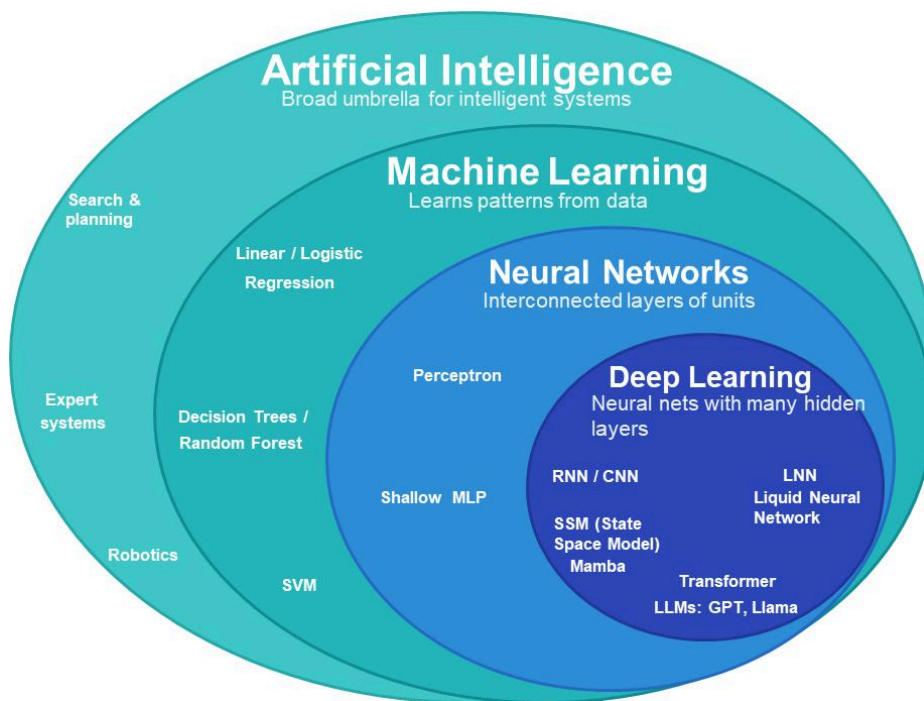
Feature	Traditional Programming	Machine Learning	Deep Learning
Rules	Manual	Learned	Automatically learned
Data Requirement	Low	Moderate	Very High
Hardware	CPU	CPU/GPU	GPU/TPU
Accuracy	Limited	High	Very High
Feature Engineering	Manual	Partial	Mostly Automatic
Best For	Simple Logic	Prediction	Complex AI Tasks

Diagrams, Tables & Visual Learning



CONVOLUTIONAL NEURAL NETWORK (CNN) FOR IMAGE CLASSIFICATION





Deep Learning Architecture

Layer	Function
Input Layer	Receives data
Hidden Layer 1	Detects simple patterns
Hidden Layer 2	Learns complex features
Hidden Layer 3	Combines representations

Layer	Function
Output Layer	Produces predictions

Popular Neural Network Types

Model	Primary Application
CNN	Image Processing
RNN	Sequential Data
LSTM	Time Series
Transformer	NLP
Autoencoder	Compression
GAN	Image Generation

Training Pipeline

Stage	Purpose
Collect Data	Build dataset
Clean Data	Improve quality
Train Model	Learn patterns
Validate	Prevent overfitting
Test	Evaluate performance
Deploy	Production use

Examples

Example 1 — Image Classification

Input:

A photograph of a cat.

Output:

```
1 | Cat
2 | Confidence = 99.2%
```

Example 2 — Medical Diagnosis

Input:

Chest X-ray

Output:

```
1 | Possible Pneumonia
2 | Confidence = 97%
```

Example 3 — Speech Recognition

Input:

Human voice

Output:

```
1 | Converted into text
```

Example 4 — Autonomous Vehicles

Input:

Road camera

Output:

- Detect pedestrians
- Detect vehicles

- Recognize traffic lights
- Predict movement

Real World Applications

Deep learning is revolutionizing multiple engineering sectors.

Healthcare

Applications include:

- Disease detection
- MRI analysis
- Drug discovery
- Personalized medicine

Manufacturing

Used for:

- Predictive maintenance
- Defect detection
- Quality inspection
- Industrial robotics

Civil Engineering

Applications:

- Crack detection
- Structural monitoring
- Bridge inspection
- Earthquake prediction research

Electrical Engineering

Deep learning assists with:

- Smart grids
- Energy forecasting
- Fault detection

- Load balancing

Mechanical Engineering

Examples:

- Predictive maintenance
- Digital twins
- Robot control
- Equipment optimization

Aerospace

Applications include:

- Aircraft inspection
- Flight optimization
- Satellite imagery analysis
- Navigation systems

Finance

Used for:

- Fraud detection
- Risk analysis
- Credit scoring
- Algorithmic trading

Common Mistakes

Many beginners encounter avoidable problems.

Using Too Little Data

Small datasets often cause poor generalization.

Ignoring Data Quality

Poor labels create inaccurate models.

Overfitting

The model memorizes training data rather than learning patterns.

Poor Hyperparameter Selection

Incorrect learning rates can prevent convergence.

Skipping Validation

Testing only on training data produces misleading accuracy.

Ignoring Model Explainability

Engineers should understand why predictions occur.

Challenges & Solutions

Challenge	Solution
Limited Data	Data augmentation
Slow Training	GPU acceleration
Overfitting	Dropout and regularization
High Cost	Cloud computing
Large Models	Model compression
Bias	Balanced datasets
Low Accuracy	Better preprocessing

Case Study

Smart Factory Defect Detection

A manufacturing company struggled with manual inspection of electronic components.

Problem

- Human fatigue
- Missed defects
- High labor costs

Solution

A CNN model was trained using over 500,000 labeled images.

The system learned to detect:

- Scratches
- Missing components
- Solder defects
- Broken connectors

Results

✓ Inspection speed increased by 400%

✓ Detection accuracy reached 99%

📊 Production waste decreased

✓ Customer satisfaction improved

This demonstrates how deep learning delivers measurable engineering value.

Essential Tips

Build Strong Mathematics

Focus on:

- Linear Algebra
- Probability
- Calculus

- Statistics

Learn Python

Python remains the dominant programming language for AI.

Useful libraries include:

- NumPy
- Pandas
- Matplotlib
- TensorFlow
- PyTorch

Practice Regularly

Theory alone is insufficient.

Implement projects involving:

- Image classification
- Object detection
- NLP
- Time-series forecasting

Understand the Data

Better data often beats larger models.

Monitor Performance

Track:

- Loss
- Accuracy
- Validation metrics
- Resource usage

Continue Learning

Deep learning evolves rapidly, making continuous education essential.

Frequently Asked Questions ?

1. Is this book suitable for beginners?

Yes. It introduces deep learning concepts progressively while also covering advanced topics for experienced practitioners.

2. Do I need advanced mathematics?

Basic knowledge of algebra, probability, and calculus is helpful, but many practical examples can be followed with only a foundational understanding.

3. Which [programming language](#) is used?

Python is the primary language because of its rich AI ecosystem.

4. Is GPU hardware required?

Not initially. Beginners can start on CPUs, while GPUs significantly reduce training time for larger models.

5. Which industries use deep learning?

Healthcare, manufacturing, automotive, aerospace, finance, cybersecurity, retail, telecommunications, and scientific research all benefit from deep learning.

6. What is the biggest advantage of deep learning?

Its ability to automatically learn complex representations from large datasets, reducing the need for manual feature engineering.

7. How long does it take to learn deep learning?

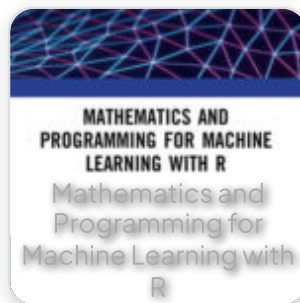
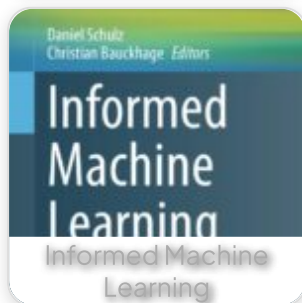
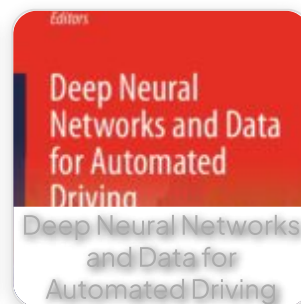
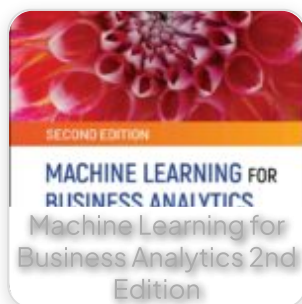
With consistent study and hands-on practice, learners can build a strong foundation in a few months, while mastering advanced architectures and deployment typically takes longer.

Conclusion

Fundamentals of Deep Learning, 2nd Edition provides a comprehensive pathway into one of the most transformative technologies of the modern era. By combining theoretical principles with practical implementation, it equips readers to design, train, evaluate, and deploy neural networks that solve complex engineering challenges.

Whether your goal is to build intelligent robots, develop medical diagnostic systems, optimize manufacturing processes, or create next-generation AI applications, the concepts presented in this guide form a solid foundation. As deep learning continues to shape industries worldwide, investing time in understanding these fundamentals will prepare students, researchers, and professionals for the rapidly evolving future of artificial intelligence.

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